Voice Arena Leaderboard: Executive One-Pager

Product Overview & Evaluation Framework for Speaker Diarization

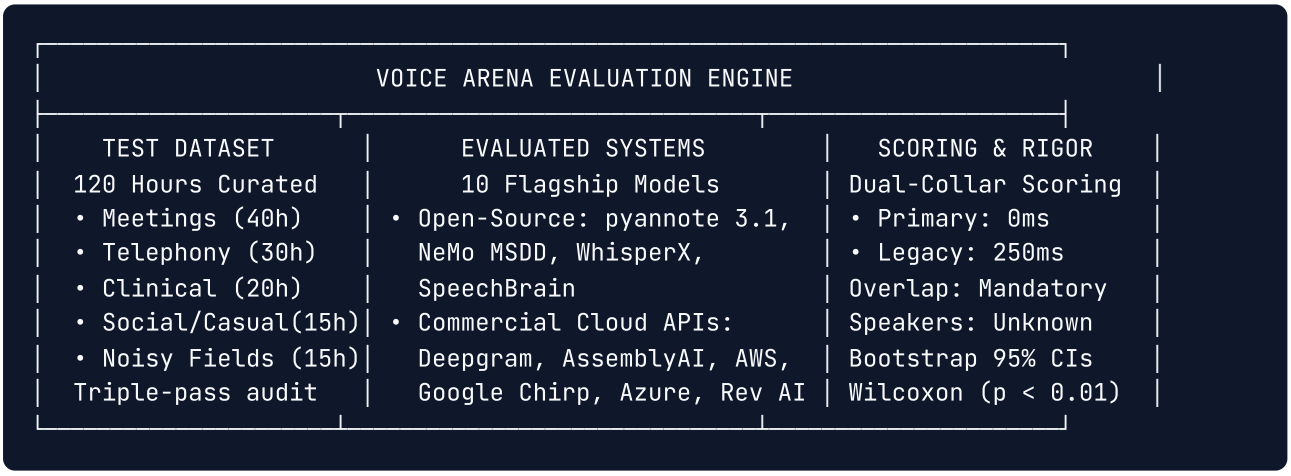
Target Reviewer: Executive Team & Engineering Leadership | **Read Time:** 2 Minutes

The Problem & Market Need

Speaker diarization ("who spoke when") is essential for meeting intelligence, contact centers, and clinical scribes. Yet published vendor accuracy numbers are systematically misleading: they rely on sterile academic datasets, forgive boundary errors with generous 250ms collars, and ignore cross-talk overlap. Product teams select systems based on marketing claims only to encounter severe conversational turn scrambling in production.

Voice Arena's Mission: Provide the definitive, unbiased, production-grounded benchmark for speech diarization systems.

Core Leaderboard Architecture



The 4 Crucial Design Choices

DESIGN CHOICE	RATIONALE & TRADE-OFF
1. Strict 0ms Boundary Collar	Traditional 250ms collars erase half a second of error per speaker turn, hiding speech collisions. 0ms reflects true production turn-taking.
2. Mandatory Overlap Evaluation	12% of real conversation is cross-talk, driving 60%+ of downstream errors. We score overlap without forgiveness and report dedicated Overlap-DER.
3. JER Alongside DER	Diarization Error Rate is time-weighted and hides failures on quiet participants. Jaccard Error Rate treats all speakers equally, preventing conversational bias.
4. Paired Statistical Significance	Benchmark scores often differ by fractions of a percent. We enforce 1,000 bootstrap resamples and Wilcoxon significance flags before declaring a winner.

Leaderboard Output: What Decision-Makers See

Next to each ranked model, the leaderboard publishes: 1. **DER (0ms & 250ms)** with $\pm 95\%$ confidence bounds. 2. **Overlap Diarization Error Rate (OV-DER)** to gauge multi-speaker robustness. 3. **Operational Metrics:** Real-Time Factor (RTF), Streaming Latency, and Cost per 1,000 audio hours. 4. **Domain Breakdown Radar:** Instant filtering by Enterprise Meetings, Telephony, or Medical.

Key Takeaway for Voice AI Teams

> ***A model with an 8% DER on clean benchmark audio can collapse to 35% DER in real multi-speaker meetings.*** *Voice Arena eliminates blind vendor selection by pairing realistic acoustic stress-tests with statistical confidence intervals and cost-latency profiling.*